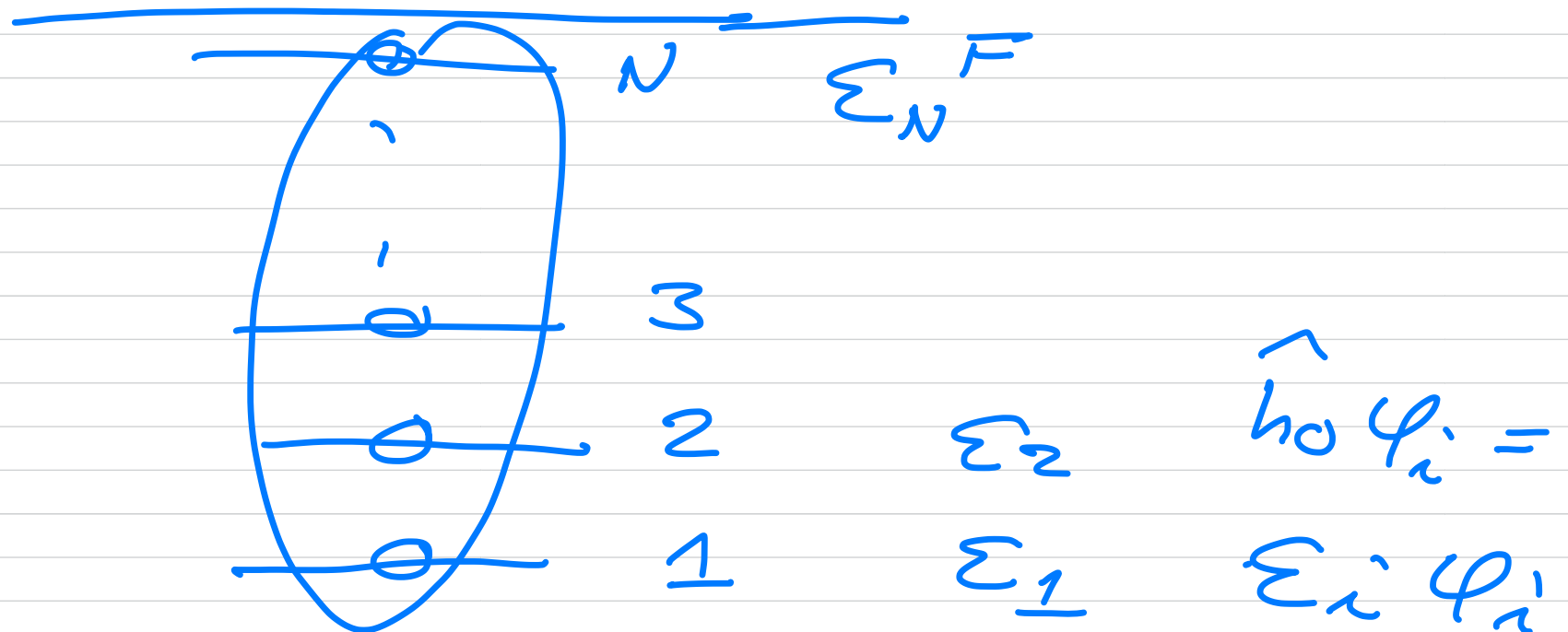
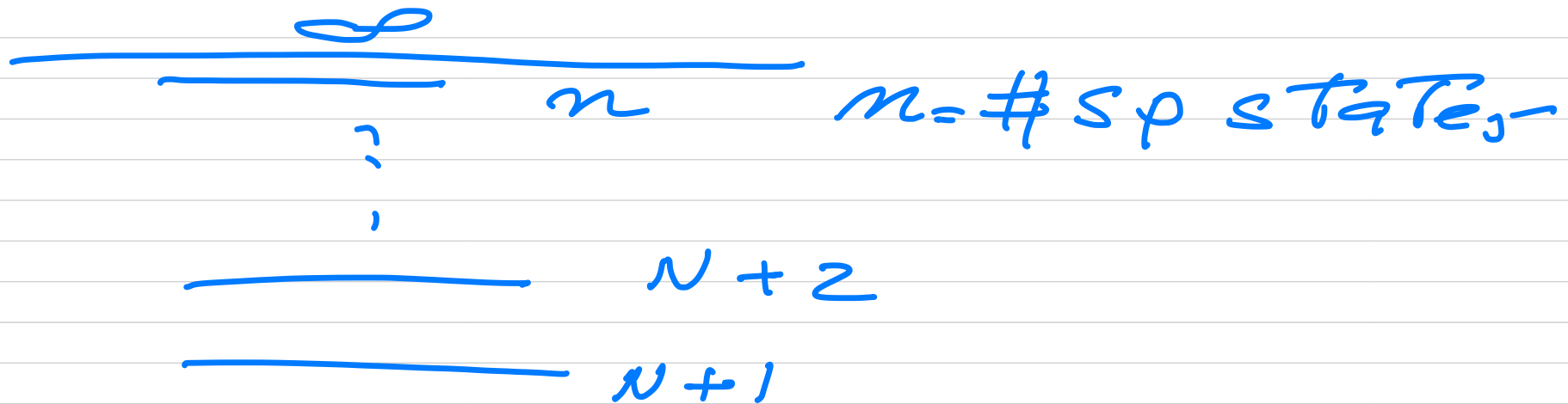


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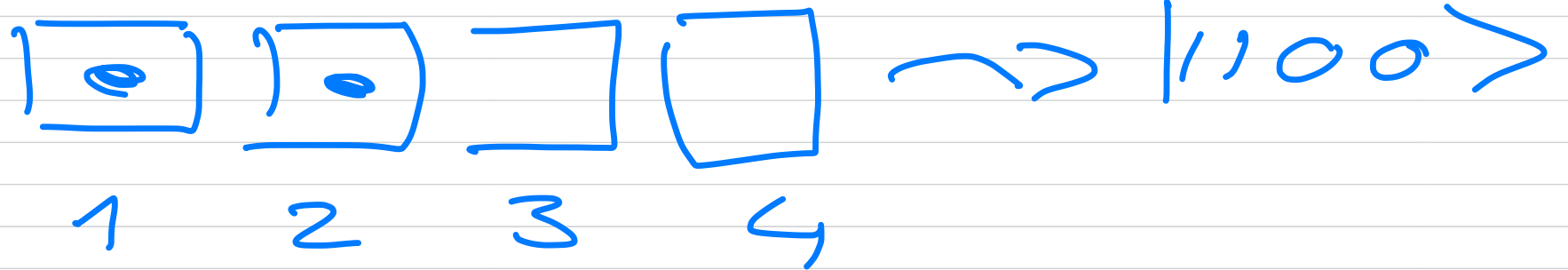


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$$N = 2$$

$$n = 4$$



vacuum state $|0\rangle$

Define an operator (creation)

$$a_p^+ |0\rangle = |p\rangle$$

$$|1\rangle = a_1^+ |0\rangle$$

$$|12\rangle = a_1^+ a_2^+ |0\rangle$$

$$a_p^\dagger |p\rangle = a_p^\dagger a_p^\dagger |0\rangle = 0$$

$$|11\rangle = 0$$

$$a_p^\dagger |123 \dots N\rangle$$

$$a_p^\dagger a_1^\dagger a_2^\dagger a_3^\dagger \dots a_N^\dagger |0\rangle \neq 0$$

$$\left(\begin{array}{l} \text{only if } p \notin \{1, 2, 3, \dots, N\} \\ = |p123 \dots N\rangle \end{array} \right.$$

$$\underbrace{|12\rangle}_{SD} = a_1^\dagger a_2^\dagger |0\rangle$$

$$|12\rangle = -|21\rangle$$

$$a_1^\dagger a_2^\dagger |0\rangle = -a_2^\dagger a_1^\dagger |0\rangle$$

$$a_1^\dagger a_2^\dagger = -a_2^\dagger a_1^\dagger \Rightarrow$$

$$a_1^\dagger a_2^\dagger + a_2^\dagger a_1^\dagger = 0 \Rightarrow$$

$$\{a_1^\dagger, a_2^\dagger\} = 0 \quad ([a_1^\dagger, a_2^\dagger])$$

$$\{a_p^\dagger, a_q^\dagger\} = 0$$

Define annihilation/destruction
operator

$$a_p = (a_p^\dagger)^\dagger$$

$$a_p |0\rangle = 0$$

$$a_p |p\rangle = |0\rangle$$

$$\{a_p, a_q\} = 0 = [a_p, a_q]_+$$

$$|12\rangle = a_1^\dagger a_2^\dagger |0\rangle \quad (N=2)$$

$$a_p |12\rangle \quad p \in \{1, 2\}$$

$$\begin{aligned} a_1 |12\rangle &= a_1 a_1^\dagger a_2^\dagger |0\rangle \\ &= a_2^\dagger |0\rangle = |2\rangle \\ &\quad (N=1) \end{aligned}$$

Mixed commutator :

$$\{a_p, a_q^\dagger\} = \delta_{pq}$$

$$\langle p | q \rangle$$

$$\langle p | = \langle 0 | a_p$$

$$| p \rangle = a_p^\dagger | 0 \rangle$$

$$\begin{aligned} \langle p | q \rangle &= \langle 0 | a_p a_q^\dagger | 0 \rangle \\ &= \delta_{pq} \end{aligned}$$

$$\langle 0 | a_p^\dagger a_q | 0 \rangle = 0$$

$$a_p a_q^\dagger + a_q^\dagger a_p = \delta_{pq}$$

For fermions

$$\{a_p^\dagger, a_q^\dagger\} = \{a_p, a_q\} = 0$$

$$\{a_p, a_q^\dagger\} = \delta_{pq}$$

$$N \geq 2$$

$$|12 \dots k \dots \alpha \dots k+1 \dots N\rangle$$

$$= a_1^+ a_2^+ \dots a_k^+ a_\alpha^+ a_{k+1}^+ \dots a_N^+ |0\rangle$$

$$a_p^+ a_p |12 \dots k \dots \alpha \dots k+1 \dots N\rangle$$

$$p \in \{1, 2, \dots, N\}$$

$$a_\alpha^+ a_\alpha a_1^+ a_2^+ \dots a_k^+ a_\alpha^+ a_{k+1}^+ \dots a_N^+ |0\rangle$$

$$(-1)^k a_\alpha^\dagger \underbrace{|12 \dots k k+1 \dots N\rangle}_{N-1 \text{ state}}$$

$$(-1)^k |\alpha 12 \dots k k+1 \dots N\rangle_{N \text{ state}}$$

$$(-1)^{2k} |12 \dots k \alpha k+1 \dots N\rangle = |12 \dots k \alpha k+1 \dots N\rangle$$

$$a_p a_p^\dagger |12 \dots k \alpha k+1 \dots n\rangle$$

$$p \in \{1, 2, \dots, k \alpha k+1, n\}$$

then zero

$$a_p |p 12 \dots k \alpha k+1 \dots n\rangle$$

$n+1$ state

$$p > n$$

$$\begin{aligned} (-1)^N a_p |12 \dots k \dots n p\rangle \\ = |12 \dots k \dots n\rangle \end{aligned}$$

$$a_p a_q^\dagger |12 \dots n\rangle$$

$$a_p^\dagger a_q |12 \dots n\rangle$$

$$\langle \underline{\Phi}_1(x_1, x_2) | \underline{\Phi}_2(x_1, x_2) \rangle$$

$$= \frac{1}{2} \int dx_1 \int dx_2 (\varphi_p^*(x_1) \varphi_q^*(x_2) - \varphi_p^*(x_2) \varphi_q^*(x_1))$$

$$\times (\varphi_2(x_1) \varphi_5(x_2) - \varphi_2(x_2) \varphi_5(x_1))$$

$$\langle \varphi_p | \varphi_q \rangle = \delta_{pq}$$

$$= \delta_{rp} \delta_{sq} - \delta_{ps} \delta_{qr}$$

$$\underline{\Phi}_2(x_1, x_2) \rightarrow a_r^\dagger a_s^\dagger |0\rangle$$

$$\underline{\Phi}_1(x_1, x_2) \rightarrow \overset{= |rs\rangle}{a_p^\dagger a_q^\dagger |0\rangle}$$

$$= |pq\rangle$$

$$\langle p | q \rangle = \langle 0 | a_p a_q^\dagger | 0 \rangle$$

$$= \langle 0 | \delta_{pq} - a_q^\dagger a_p | 0 \rangle$$

$$\langle p q | r s \rangle =$$

$$\langle 0 | a_q a_p a_r^\dagger a_s^\dagger | 0 \rangle$$

$$a_q a_p a_r^\dagger a_s^\dagger =$$

$$a_q (\delta_{pr} - a_r^\dagger a_p) a_s^\dagger$$

$$a_q a_s^\dagger \delta_{pr} - \boxed{a_q a_r^\dagger a_p a_s^\dagger}$$

$$\boxed{(\delta_{qs} - a_s^\dagger a_q) \delta_{pr}}$$

$$= \delta_{qs} \delta_{pr} \checkmark - \delta_{pr} a_s^\dagger a_q$$

$\langle 0 | a_s^\dagger a_q | 0 \rangle = 0$

$$- a_q a_n^\dagger a_p a_s^\dagger$$

$$= - \boxed{a_q a_n^\dagger (\delta_{sp} - a_s^\dagger a_p)}$$

$$- a_q a_n^\dagger \delta_{sp}$$

$$- (\delta_{qn} - a_n^\dagger a_q) \delta_{sp}$$

$$- \delta_{qn} \delta_{sp} + \delta_{sp} a_n^\dagger a_q$$

Motivate Wick's theorem

- contraction
- Normal ordering
- Definition of contractions

$$\langle 0 | a_p a_q^\dagger | 0 \rangle = \delta_{pq} = \overbrace{a_p a_q^\dagger}^+$$

$$\begin{aligned} &= \langle p | q \rangle \\ \overbrace{a_p a_q}^+ &= 0 = \overbrace{a_p^\dagger a_q^\dagger}^+ \\ \overbrace{a_p^\dagger a_q}^+ &= \langle 0 | a_p^\dagger a_q | 0 \rangle = 0 \end{aligned}$$

$$\langle 0 | a_p^\dagger a_q^\dagger | 0 \rangle \quad a_p | 0 \rangle$$

Normal ordering

$$N[x y z \dots w] =$$

$$(-)^K \langle 0 | x' y' z' \dots w' | 0 \rangle$$

all creation operators to the left and all annihilation operators to the right

$K = \#$ times we need to swap operators

$$\langle p|q \rangle = \langle 0|a_p a_q^\dagger|0 \rangle$$

$$= \langle 0|\delta_{pq}|0 \rangle$$

$$- \langle 0|a_q^\dagger a_p|0 \rangle$$

$$\{a_p, a_q^\dagger\} = \delta_{pq}$$

$$= a_p a_q^\dagger$$

$$+ N[a_p a_q^\dagger]$$

$$N[a, q_2 q_3^+] \quad 1 \neq 2 \neq 3$$

$$k=2 \quad = \langle c | q_3^+ a, q_2 | c \rangle = 0$$

$$\langle 25 | p q \rangle = \langle c | a_s q_2 q_p^+ q_q^+ | 0 \rangle =$$

$$\delta_{rp} \delta_{qs} - \delta_{rp} q_q^+ q_s$$

$$N[a_s q_2 a_p^+ a_q^+] + N[a_s q_2 q_p^+ q_q^+]$$

$$- a_s q_p^+ q_2 q_q^+$$

$$\begin{aligned}
 & - a_s a_p^\dagger (\delta_{17} - a_7^\dagger a_2) = \\
 & + a_s a_p^\dagger a_7^\dagger a_2 - \delta_{17} a_s a_p^\dagger
 \end{aligned}$$

$$- \delta_{17} (\delta_{sp} - a_p^\dagger a_s)$$

$$N \left[\overbrace{a_s a_2 a_p^\dagger a_7^\dagger} \right] + N \left[\overbrace{a_s a_2 a_p^\dagger a_7^\dagger} \right]$$

$$= \langle 0 | a_n a_s a_p^\dagger a_q^\dagger | 0 \rangle$$

$$= \underbrace{N [a_n a_s a_p^\dagger a_q^\dagger]}$$

$$+ \sum_{\text{all pairs of}} N [a_n a_s a_p^\dagger a_q^\dagger]$$

of
1 set of contractions

$$+ \sum_{\text{all operators contracted}} N [a_n a_s a_p^\dagger a_q^\dagger]$$

operators
contracted